

Lab Exercise 6

Snort IDS/IPS to detect and prevent real time threats in TryHackMe Platform

AIM:

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE REQUIRED:

TryHackMe account, AttackBox or VPN, Snort, Nmap, Linux terminal.

HOW TO START TRYHACKME

1. Login to TryHackMe.
2. Search for "Snort" room.
3. Click Start Machine.
4. Click Start AttackBox.
5. Open Terminal in AttackBox.
6. Verify Snort:

```
snort -V
```

7. Test configuration:

```
sudo snort -T -c /etc/snort/snort.conf
```

Experiment 1 — ICMP (Ping) Detection (IDS) — line by line (with new terminal steps)

1. Open terminal and Check Snort:

```
snort -V
```

2. Test Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

3. Open rule File

```
sudo nano /etc/snort/rules/local.rules
```

4. Add this rule:(ENSURE ONLY THIS RULE EXISTS)

```
alert icmp any any -> any any (msg:"ICMP Ping Detected";  
sid:1000001; rev:1;)
```

5. Save and exit: Ctrl+O → Enter → Ctrl+X

6. Run Snort (IDS Mode)

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

7. Generate Attack (New Terminal) using Open new terminal

In AttackBox:

- Click + (New Tab) at top of terminal
OR

- Right-click terminal → New Tab

8. In the new terminal:(ATTACK BOX IP ADDRESS)

```
ping 10.48.190.168
```

9. Go back to first terminal and View Alert → confirm alert shown
ICMP Ping Detected

10.(Log check – optional):

```
sudo tail -f /var/log/snort/alert
```

```
ubuntu@ip-10-48-190-168:~$ ping 10.48.190.168  
PING 10.48.190.168 (10.48.190.168) 56(84) bytes of data.  
64 bytes from 10.48.190.168: icmp_seq=1 ttl=64 time=0.020 ms  
64 bytes from 10.48.190.168: icmp_seq=2 ttl=64 time=0.042 ms  
64 bytes from 10.48.190.168: icmp_seq=3 ttl=64 time=0.032 ms  
64 bytes from 10.48.190.168: icmp_seq=4 ttl=64 time=0.038 ms  
64 bytes from 10.48.190.168: icmp_seq=5 ttl=64 time=0.044 ms  
64 bytes from 10.48.190.168: icmp_seq=6 ttl=64 time=0.040 ms
```

```
64 bytes from 10.48.190.168: icmp_seq=412 ttl=64 time=0.039 ms
^C
--- 10.48.190.168 ping statistics ---
412 packets transmitted, 412 received, 0% packet loss, time 420857ms
rtt min/avg/max/mdev = 0.005/0.037/0.048/0.004 ms
ubuntu@ip-10-48-190-168:~$
```

```
Commencing packet processing (pid=1941)
S5: Session exceeded configured max segs to queue 2621 using 2621 segs (server queue). 10.48.127.64 4226
4 --> 10.48.190.168 80 (0) : LWstate 0x40 LWFlags 0x2101
03/10-10:22:44.966713  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::79:32f
f:fe6b:91a7 -> ff02::2
```

Experiment 2 — Port Scan Detection — line by line (with new terminal steps)

1. Edit Rules File

```
sudo nano /etc/snort/rules/local.rules
```

2. Add:

```
alert tcp any any -> any any (msg:"TCP Port Scan Activity";
sid:1000002; rev:1;)
```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

```
Ctrl + C
```

5. Restart Snort:

```
sudo snort -A console -c /etc/snort/snort.conf -i eth0
```

6. Open new terminal tab

- Click + on terminal
OR
- Right-click → New Tab

7. In new terminal:

```
for p in {1..100}; do nc -zv 10.48.190.168 $p; done
```

8. Go back to Snort terminal → confirm alert

9. (Optional log):

```
sudo tail -f /var/log/snort/alert
```

```
03/10-11:03:14.096390  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096939  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096994  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.098369  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.099898  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.099989  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.101333  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118446  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118519  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.119767  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2264 -> 10.48.190.168:80
```

```
nc: connect to 10.48.190.168 port 14 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 15 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 16 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 17 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 18 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 19 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 20 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 21 (tcp) failed: Connection refused
Connection to 10.48.190.168 22 port [tcp/ssh] succeeded!
nc: connect to 10.48.190.168 port 23 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 24 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 25 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 26 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 27 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 28 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 29 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 30 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 31 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 32 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 33 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 34 (tcp) failed: Connection refused
```

Experiment 3 — IPS Blocking (Inline Mode) — line by line (with new terminal steps)

1. Change rule in Rule File

```
sudo nano /etc/snort/rules/local.rules
```

2. Add:

```
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003;  
rev:1;)
```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

Ctrl + C

5. Run inline:

```
sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf
```

6. Open new terminal tab

- Click +

OR

- Right-click → New Tab

7. In new terminal:

```
ping 8.8.8.8
```

8. Expected result:

 No ping reply

✓ Alert shown in Snort terminal

9. (Optional log):

```
sudo tail -f /var/log/snort/alert
```

If interface is not eth0

Run once:

ip a

Use the shown interface name instead of eth0.

RESULT:

Successfully detected and prevented attacks using Snort IDS/IPS.